

Institute of Actuaries of India

Subject CT1 – Financial Mathematics

November 2011 Examinations

INDICATIVE SOLUTIONS

Note

- 1. Markers are instructed to follow the guidelines as closely as possible while giving step marks**
- 2. No step marks need to be given for answers lacking fundamental understanding of the concepts**
- 3. Any alternative solution should be given full credit provided it is logical**
- 4. Any incorrect solution should not be penalized more than once**

Sol. 1)**a)**

The amount initially lent was $50,000(1 - \frac{9}{12} * 0.18) = 43,250$

b)**i.**

$$(1+i) = e^{\delta} = 1.0618$$

$$v = 0.9418$$

$$d^{(12)} = 12(1 - v^{(1/12)}) = 5.98\%$$

ii.

$$v = 1 - d = \left(1 - \frac{d^{(4)}}{4}\right)^4 = 0.9413$$

$$i = v^{-1} - 1 = 6.24\%$$

$$i^{(12)} = 12 \left((1+i)^{\frac{1}{12}} - 1 \right) = 6.07\%$$

c)

The order is determined based on when the interest is paid, so, for example d corresponds to interest paid immediately which requires a smaller payment amount. Similarly, i corresponds to interest paid at the end of time and δ corresponds to payment throughout

$$d < \delta < i^{(4)} < i$$

d)

With 1 unit of initial capital, d is the interest payable at the beginning of the year. Similarly, with the same capital, i is the interest payable at the end of the year. So d must be equal to the present value at the beginning of the year of the payment i payable at the end of the year. i.e. $d = iv$

[10]

Sol. 2)

a)

$i^{(12)} = 6\%$, thus monthly effective i is given by

$$i = \frac{i^{(12)}}{12} = 0.5\%$$

Accumulation factor for two years (working in months) = $(1 + 0.5\%)^{24} = 1.1272$

Accumulated amount after two years = 7945.63

Accumulation factor for next 2.5 years (working in quarters) = $(1 + 10 \times 1.5\%) = 1.15$

Accumulated amount after four and a half years = 9137.48

$$v = 1 - d = \left(1 - \frac{d^{(12)}}{12}\right)^{12} = 0.94162$$

$$i = v^{-1} - 1 = 6.2\%$$

Accumulation factor for next 1.5 years (working in years) = $(1 + 6.2\%)^{1.5} = 1.0944$

Accumulated amount after 6 years = $7,049 \times 1.1272 \times 1.15 \times 1.0944 = 10,000$

b)

Present value of variable annuity PV is given by

$$PV = 1^2 + 2^2v + 3^2v^2 + 4^2v^3 + \dots + n^2v^{(n-1)}$$

$$\Rightarrow PV = 1 + 4v + 9v^2 + 16v^3 + \dots + n^2v^{(n-1)}$$

$$\Rightarrow vPV = v + 4v^2 + 9v^3 + 16v^4 + \dots + (n-1)^2v^{(n-1)} + n^2v^n \text{ (multiplying both sides by } v)$$

subtracting,

$$(1-v)PV = 1 + 3v + 5v^2 + 7v^3 + \dots + (2n-1)v^{(n-1)} + n^2v^n \text{ (since } n^2 - (n-1)^2 = (n+n-1) \times (n-n+1) = 2n-1)$$

$$\Rightarrow (1-v)PV = 2\{1 + 2v + 3v^2 + 4v^3 + \dots + nv^{(n-1)}\} - \{1 + v + v^2 + v^3 + \dots + v^{(n-1)}\} + n^2v^n$$

$$PV = \frac{2(I\ddot{a}_{\overline{n}|}) - \ddot{a}_{\overline{n}|} - n^2v^n}{1-v}$$

[8]

Sol.3)

a)

The first two conditions of Redington's theory of immunization

- present value of assets and liabilities should be equal
- volatility of assets and liabilities should be equal

The risk free rate of return (i) is 6% per annum effective.

Present value of liabilities ($V_L(i)$)

$$x * 1.06^{-1} + 2x * 1.06^{-2} + 3.5x * 1.06^{-3} + 4.5x * 1.06^{-4} + 5x * 1.06^{-5} + 100 * 1.06^{-6}$$

$$= 12.9628x + 70.4961$$

Present value of assets ($V_A(i)$)

$$10 * 1.06^{-1} + 10 * 1.06^{-2} + 20 * 1.06^{-5} + 30 * 1.06^{-6} + y * 1.06^{-8}$$

$$= 54.4279 + 0.6274y$$

As per Redington's first condition

Present value of liabilities = Present value of assets

$$\Rightarrow 12.9628x + 70.4961 = 54.4279 + 0.6274y$$

$$\Rightarrow x = 0.0484y - 1.2396$$

Volatility of liabilities is $-V'_L(i)/V_L(i)$, where $-V'_L(i)$ is

$$x * 1.06^{-2} + 2 * 2x * 1.06^{-3} + 3 * 3.5x * 1.06^{-4} + 4 * 4.5x * 1.06^{-5}$$

$$+ 5 * 5x * 1.06^{-6} + 6 * 100 * 1.06^{-7}$$

$$= 43.6401x + 399.0343$$

Volatility of assets is $-V'_A(i)/V_A(i)$, where $-V'_A(i)$ is

$$10 * 1.06^{-2} + 2 * 10 * 1.06^{-3} + 5 * 20 * 1.06^{-6} + 6 * 30 * 1.06^{-7} + 8 * y * 1.06^{-9}$$

$$= 215.8987 + 4.7352y$$

As per Redington's second condition

Volatility of liabilities = Volatility of assets

$$\Rightarrow 43.6401x + 399.0343 = 215.8987 + 4.7352y, \text{ since } V_L(i) = V_A(i) \text{ as per first condition}$$

$$\Rightarrow 43.6401 * (0.0484y - 1.2396) + 399.0343 = 215.8987 + 4.7352y$$

$$\Rightarrow y = \frac{129.041}{2.623}$$

$$\Rightarrow y = 49.1967$$

Hence

$$x = 0.0484 * 49.1946 - 1.2396 = 1.1416$$

(The second condition can also be taken as equality of DMTs and solved.)

Convexity of liabilities is $V''_L(i)/V_L(i)$, where $V''_L(i)$ is

$$2 * 1.1416 * 1.06^{-3} + 2 * 3 * 2 * 1.1416 * 1.06^{-4} + 3 * 4 * 3.5 * 1.1416 * 1.06^{-5} \\ + 4 * 5 * 4.5 * 1.1416 * 1.06^{-6} + 5 * 6 * 5 * 1.1416 * 1.06^{-7} + 6 * 7 * 100 * 1.06^{-8}$$

b)

$$= 2,870.047$$

$$V_L(i) = V_A(i) = 12.9628 * 1.1416 + 70.4961 = 85.294$$

$$\text{Hence, convexity of liabilities} = 2,870.047 / 85.294 = 33.649$$

Convexity of assets is $V''_A(i)/V_A(i)$, where $V''_A(i)$ is

$$2 * 10 * 1.06^{-3} + 2 * 3 * 10 * 1.06^{-4} + 5 * 6 * 20 * 1.06^{-7} + 6 * 7 * 30 * 1.06^{-8} \\ + 8 * 9 * 49.1946 * 1.06^{-10}$$

$$= 3,231.819$$

$$\text{Hence, convexity of assets} = 3,231.819 / 85.294 = 37.890$$

Here convexity of assets > convexity of liabilities, which is the third condition for Redington's immunization theory.

As all the conditions for Redington's immunization theory are satisfied the portfolio is immunized against small changes in interest rate.

[12]

Sol. 4)**a)**

Effective rate of return, $i = (1 - d)^{-1} - 1 = 6.383\%$

The purchase price of the house P is given by,

$$P = 5,000 * 12 * \ddot{a}_{\overline{1}|}^{(12)} (1 + 1.052v + 1.052^2 v^2 + 1.052^3 v^3 + \dots) \\ - 2,000(1 + 1.02v + 1.02^2 v^2 + 1.02^3 v^3 + \dots)$$

$$\text{Where, } 12 * \ddot{a}_{\overline{1}|}^{(12)} = 12 * 1.06383^{(1/12)} * \frac{1 - 1.06383^{-1}}{12 * (1.06383^{(1/12)} - 1)} = 11.66631$$

$$= 5,000 * 11.66631 * \bar{a}_{\infty}^{-j} - 2,000 * \bar{a}_{\infty}^{-k}$$

$$\text{Where } j = \frac{1.06383}{1.052} - 1 = 0.011245 \text{ and } k = \frac{1.06383}{1.02} - 1 = 0.042970$$

$$= 5,000 * 11.66631 * \frac{1+j}{j} - 2,000 * \frac{1+k}{k}$$

$$= 51,97,101$$

b)

The monthly rent of the house on 1st April, 2011 will be

$$= 5,000 * 1.052^2 = 5,533.52$$

The maintenance cost of the house on 1st April, 2011 will be

$$= 2,000 * 1.02^2 = 2,080.80$$

The sale price S of the house as on 1st April, 2011 is given by,

$$S = 5,533.52 * 12 * \ddot{a}_{\overline{1}|}^{(12)} (1 + 1.053v + 1.053^2 v^2 + 1.053^3 v^3 + \dots) \\ - 2,080.80(1 + 1.02v + 1.02^2 v^2 + 1.02^3 v^3 + \dots)$$

$$= 5,533.52 * 11.66631 * \bar{a}_{\infty}^{-j} - 2,080.80 * \bar{a}_{\infty}^{-k}$$

$$\text{Where } j = \frac{1.06383}{1.053} - 1 = 0.0102847 \text{ and } k = \frac{1.06383}{1.02} - 1 = 0.042970$$

$$= 5,533.52 * 11.66631 * \frac{1+j}{j} - 2,080.80 * \frac{1+k}{k}$$

$$= 62,90,927$$

Calculation for capital gain tax

Purchase price of property after inflation allowance

$$= 51,97,101 * \frac{106}{99} = 55,64,573$$

$$\text{Capital gain tax} = 10\% * (62,90,927 - 55,64,573) = 72,635$$

Amount received on sale of property after capital gain tax

$$= 62,90,927 - 72,635 = 62,18,292$$

[10]

Sol. 5)

- (i) Merits and Demerits of Index-linked government bond:
- + Government bonds are more secured.
 - + Marketability of the security is good.
 - The underlying Whole Price Index may not match with index followed by the liability.
 - Indexation lag of 6 months means that index linked liability is not matched completely with the asset.
- (ii) Merits and Demerits of interest rate swap offered by investment bank:
- + The asset will match the liability more closely than index-linked government bond. There may not be any indexation lag.
 - There is a credit risk associated with the issuer bank. Issuer bank may default.
 - The market condition may change in future making the swap contract unviable to the customer.
 - The swap may be designed only for a particular customer. Hence, the marketability of the security is poor.
 - The cost of such swap may be more as it is specially designed for the customer. Also the issuer will keep allowance for profit margin in the contract.

[5]

Sol.6)

Par yield:

- a) The n-year par yield represents the coupon per £1 nominal that would be payable on a bond with term n years, which would give the bond a current price under the current term structure of £1 per £1 nominal, assuming the bond is redeemed at par.
- b) To find out par coupons of a four year bond issued at time t, we need to determine 1-year, 2-year, 3-year and 4-year discounting factors at time point t=2

Hence, the par coupon c of a bond with redemption amount £100, is given by

$$100 = c \left(\frac{1.055^2}{1.06^3} + \frac{1.055^2}{1.07^4} + \frac{1.055^2}{1.08^5} + \frac{1.055^2}{1.085^6} \right) + 100 \frac{1.055^2}{1.085^6}$$

$$= c(0.9345 + 0.8491 + 0.7575 + 0.6822) + 100 * 0.6822$$

(0.5 marks for correct method of discounting and 0.5 marks for correct calculation of each of the five discounting factors)

$$\Rightarrow c = \frac{100 - 68.22}{0.9345 + 0.8491 + 0.7575 + 0.6822} = 9.86$$

[6]

Sol.7)

a) Arbitrage is a risk free trading profit. Arbitrage free contract is a contract where no arbitrage opportunities exist.

An arbitrage opportunity exists if either:

(a) an investor can make a deal that would give her or him an immediate profit, with no risk of future loss, or

(b) an investor can make a deal that has zero initial cost, no risk of future loss, and a non-zero probability of a future profit.

b) i.

Total number of convertible units to be issued after 3 months = $500 * 1,000 = 5,00,000$

The cost of the underlying stocks to be issued to employees on the date of issuance i.e. at 3 month's time, $S = 5,00,000 * 15 = \text{₹} 75,00,000$

The amount of dividend due on the stocks before the employees can convert them

Time (t) of dividend payment (from now)	Amount of dividend payment (in ₹)
1 year and 3 months from now	$0.25 * 5,00,000 = 1,25,000$
2 years and 3 months from now	$0.25 * 5,00,000 * 1.02 = 1,27,500$
3 years and 3 months from now	$0.25 * 5,00,000 * 1.02^2 = 1,30,050$

On date of issuance of convertible stocks, the present value of all the dividends to be paid over next 3 years

$$I = 1,25,000 * 1.06^{-1} + 1,27,500 * 1.06^{-2} + 1,30,050 * 1.06^{-3} = 3,40,591.56$$

Hence forward price of the hedging forward contract

$$= (S - I) * 1.06^3 = (75,00,000 - 3,40,591.56) * 1.06^3 = 85,26,970$$

b) ii.

The amount the company need now to enter into the contract

$$= 85,26,970 * 1.06^{-3.25} = 70,55,872$$

[7]

- Sol. 8)** Note that the period of investigation is 3 years
Also, the cashflows received on 1 April 2008 and 1 June 2009 are not new money and are generated by the fund itself

Thus, TWRR is given by

$$(1+i)^3 =$$

$$\left(\frac{11,00,000}{10,00,000 + 0} \right) * \left(\frac{11,00,000}{11,00,000 + 3,00,000} \right) * \left(\frac{7,30,000}{11,00,000 - 2,00,000} \right) * \left(\frac{15,00,000}{7,30,000 + 300,000} \right)$$

$$= 1.1 * 0.7857 * 0.811 * 1.4563 = 1.02092$$

$$\text{TWRR} = 1.02092^{(1/3)} - 1 = 0.69\%$$

[4]

Sol. 9)

- a) The discounted payback period of an investment project is the first time at which the net present value of the cash flows from the project is positive.

- b) Considering the NPV at 8% per annum effective and working in ` Lacs

Present value of cash outflows is given by

$$25.5 \bar{a}_{\overline{4}|}^{\text{@8\%}} + 2v^4 \ddot{a}_{\overline{10}|}^{(12)\text{@8\%}} + 10v^4 (1 + 1.06v + (1.06v)^2 + \dots + (1.06v)^9)$$

$$i = 8\% \Rightarrow \delta = \log_e(1+i) = 7.696\% \text{ and } i = 8\% \Rightarrow d^{(12)} = 12(1-v^{(1/12)}) = 7.6714\%$$

$$\bar{a}_{\overline{4}|}^{\text{@8\%}} = \frac{1-v^4}{\delta} = 3.442913, \quad \ddot{a}_{\overline{10}|}^{(12)\text{@8\%}} = \frac{1-v^{10}}{d^{(12)}} = 6.997433$$

$$\text{let, } v' = 1.06v; \text{ thus } i' = v'^{-1} - 1 = 1.8868\% \text{ and } \ddot{a}_{\overline{10}|}^{\text{@}i'\%} = \frac{1-v'^{10}}{d'} = 9.206514$$

$$\text{Thus, PV of outflows} = 25.5 * 3.442913 + 2 * 0.73503 * 6.997433 + 10 * 0.73503 * 9.206514 \\ = 87.7943 + 10.2866 + 67.67063 = 165.7515$$

Present value of cash inflows is given by

$$v^4 \left(12 \bar{a}_{\overline{2}|}^{\text{@8\%}} + 17v^2 \bar{a}_{\overline{3}|}^{\text{@8\%}} + 31v^5 \bar{a}_{\overline{5}|}^{\text{@8\%}} + 120v^{10} \right)$$

$$= 0.73503 * (12 * 1.8537 + 17 * 0.85734 * 2.6789 + 31 * 0.68058 * 4.1504 + 120 * 0.46319)$$

$$= 0.73503 * (22.2444 + 39.0444 + 87.5651 + 55.5828) = 150.2671$$

Hence, NPV of the project @8% = 150.2671 – 165.7515 = -15.4844 lacs

Since NPV is negative, this means IRR is less than 8% per annum effective.

Hence, the decision criteria is not fulfilled.

[9]

Sol.10)

We will first find the accumulation factors at time $t = 3$ and $t = 6$

$$A(0,3) = e^{\int_0^3 \delta(t) dt} = e^{\int_0^3 (at+bt^2-1) dt} = e^{\left[a\frac{t^2}{2} + b\frac{t^3}{3} - t \right]_0^3} = e^{\left(\frac{9}{2}a + 9b - 3 \right)}$$

$$\ln\left(\frac{6,000}{5,000}\right) = \frac{9}{2}a + 9b - 3 = > 0.182322 = \frac{9}{2}a + 9b - 3 = > \frac{9}{2}a + 9b = 3.182322 \text{ -----(1)}$$

Similarly,

$$A(0,6) = e^{\int_0^6 \delta(t) dt} = e^{\int_0^6 (at+bt^2-1) dt} = e^{\left[a\frac{t^2}{2} + b\frac{t^3}{3} - t \right]_0^6} = e^{(18a+72b-6)}$$

$$\ln\left(\frac{10,000}{5,000}\right) = 18a + 72b - 6 = > 0.693147 = 18a + 72b - 6 = > 18a + 72b = 6.693147 \text{ -----(2)}$$

Multiplying equation (1) by 8 and subtracting from 2, we get

$$18a - 36a = -18.76543 = > a = 1.042524$$

Using the above value of a in equation 1, we have

$$b = -0.16767$$

[6]

Sol.11)

a) Monthly repayment is given by

$$\frac{13\% * 4 * 5,00,000 + 5,00,000}{12 * 4} = 15,833.33$$

b) Effective rate of interest per month i' is given by

$$15,833.33 a_{\overline{48}|}^{@i'} = 5,00,000$$

$$a_{\overline{48}|}^{@i'} = 31.57895$$

Since effective rate per annum is roughly double of flat rate (around 26% per annum taking guess,

$$i' = 2\% \Rightarrow a_{\overline{48}|}^{@i'} = 30.67312$$

$$i' = 1.9\% \Rightarrow a_{\overline{48}|}^{@i'} = 31.30675$$

Using interpolation

$$\frac{i' - 1.9\%}{2\% - 1.9\%} = \frac{31.57895 - 31.30675}{30.67312 - 31.30675} \Rightarrow i' = 1.857\%$$

$$\text{Effective rate of interest per annum} = (1 + 1.857\%)^{12} - 1 = 24.71\%$$

[5]

Sol.12) Working in months

a) The monthly installment X_1 is given by

$$X_1 a_{\overline{120}|}^{@1\%} = 10,00,000$$

$$\Rightarrow X_1 = 14,347.1$$

$$\text{Total amount paid in 5}^{\text{th}} \text{ year} = 1,72,165.2$$

b) Capital outstanding at the end of 5th year for loan = $14347.1 a_{\overline{60}|}^{@1\%} = 6,44,974.43$

Capital outstanding at the end of 4th year for loan = $14347.1a_{\overline{72}|1\%} = 7,33,859.78$

Thus, total capital repaid during the 5th year

$$= 7,33,859.78 - 6,44,974.43$$

$$= 88,885.35$$

$$\text{Total interest paid during 5th year} = 1,72,165.2 - 88,885.35 = 83,279.85$$

[8]

Sol. 13)**(i)**

The accumulated amount, S is given by

$$S = 5,000(1+i_1)(1+i_2)(1+i_3)....(1+i_{10}) + 10,000(1+i_2)(1+i_3)....(1+i_{10})$$

$$\Rightarrow S = 5,000(1+i_2)(1+i_3)....(1+i_{10})(1+i_1+2)$$

$$\Rightarrow S = 5,000(3+i_1)(1+i_2)(1+i_3)....(1+i_{10})$$

Hence the mean of accumulated amount is given by

$$E(S) = 5,000(3+j)(1+j)(1+j)....(1+j)$$

Where $j = E(i_t) = 7\%$ for any t

$$= 5,000 * (3+j)(1+j)^9$$

$$= 28,220$$

Again mean of S^2 is given by

$$E(S^2) = 5,000^2 E(9 + 6i_1 + i_1^2) E \prod_{t=2}^{10} (1+i_t)^2$$

$$= 5,000^2 (9 + 6j + j^2 + s^2) (1 + 2j + j^2 + s^2)^9$$

Where, $s^2 = \text{Var}(i_t) = (2.5\%)^2$ for any t

$$\Rightarrow E(i_t^2) = (E(i_t))^2 + \text{Var}(i_t) = j^2 + s^2$$

Hence the variance of the accumulated amount

$$\begin{aligned} \text{Var}(S) &= 5,000^2 (9 + 6j + j^2 + s^2) (1 + 2j + j^2 + s^2)^9 - (28,220)^2 \\ &= 39,74,355 \end{aligned}$$

$$SD(S) = 1,994$$

(ii)

With the fixed interest rate the accumulated value of the investment

$$\begin{aligned} &= 5,000(1 + 7.5\%)^{10} + 10,000(1 + 7.5\%)^9 \\ &= 29,478 \end{aligned}$$

As per the question,

$$S \sim N(28,220, 1,994^2)$$

Hence the required probability

$$\begin{aligned} P(S > 29,478) &= P\left(Z > \frac{29,478 - 28,220}{1,994}\right) \\ &= 1 - 0.73586 \\ &= 0.26414 \end{aligned}$$

[10]
